

STUDENT PERFORMANCE PREDICTION

Using Supervised Machine Learning

Week 6 Machine Learning Project

Dataset: UCI Student Performance Dataset

Target Variable: G3 — Final Mathematics Grade

Final Model: Random Forest Regressor

Test R^2 = 0.8020 | Test RMSE = 2.0150

1. Executive Summary

This project develops a supervised machine learning regression system for predicting students' final mathematics grade (G3) using the UCI Student Performance Dataset. The workflow covers dataset verification, data cleaning, outlier investigation, exploratory data analysis, preprocessing, model comparison, evaluation, hyperparameter tuning, final model selection, feature importance analysis and error analysis.

The final selected model is a Random Forest Regressor using the Full Information feature set, which includes the previous-period grades G1 and G2 while excluding the target G3. On the fixed held-out test set, the model achieved an MAE of 1.2005, RMSE of 2.0150 and R^2 of 0.8020.

The analysis shows that G2 is overwhelmingly the strongest predictive feature, followed by absences. The model performs substantially better for students with mid-to-high actual grades and has its largest errors among students with very low final grades.

2. Project Objectives

- Understand and validate the UCI Student Performance Dataset.
- Clean and validate the data without removing valid observations unnecessarily.
- Perform exploratory analysis to identify relationships with the final grade.
- Build reproducible preprocessing pipelines without data leakage.
- Compare multiple supervised regression algorithms.
- Evaluate models using MAE, MSE, RMSE and R^2 .
- Tune the Random Forest model using cross-validation.
- Select the strongest model based on held-out test performance.
- Analyze prediction errors and feature importance.
- Document limitations and possible future improvements.

3. Dataset Description

The primary dataset is student-mat.csv from the UCI Student Performance Dataset. It contains 395 student records and 33 columns. The project uses final mathematics grade G3 as the regression target.

Dataset Property	Value
Records	395
Columns	33
Input Features	32
Target	G3
Target Type	Numerical / Regression
Target Range	0–20
Missing Values	0
Duplicate Rows	0

4. Data Cleaning and Validation

The original UCI file is semicolon-separated and was loaded using the appropriate delimiter. The cleaned dataset is stored as a standard comma-separated CSV so that it can be consumed consistently by the downstream machine learning pipeline.

No missing values or duplicate rows were found. All logical range checks for age, parental education, travel time, study time, failures, survey variables, absences and grades passed without violations.

The dataset contains 38 observations where G3 equals zero. These observations were retained because zero is within the valid grade range and the investigation did not provide evidence that they were invalid.

The IQR method identified 15 extreme absence observations above the upper threshold of 20. These were retained because an IQR flag indicates statistical extremeness rather than confirmed data-entry error. The difference in mean G3 between the normal-absence and extreme-absence groups was only about 0.154 points.

Metric	Value
Initial rows	395
Final rows	395
Rows removed	0
Missing values	0
Duplicate rows	0
G3 = 0 records	38
Absence IQR candidates	15

5. Exploratory Data Analysis

Exploratory analysis was performed to understand the target distribution and identify relationships between academic performance, attendance, study behavior and demographic or educational variables.

The final grade G3 has a mean of 10.415, a median of 11 and a standard deviation of 4.581. The distribution has a negative skew of approximately -0.733.

The strongest linear relationships with G3 are observed for G2 (correlation ≈ 0.9049) and G1 (correlation ≈ 0.8015). Failures show a negative relationship with G3 (≈ -0.3604). Absences have only a weak linear correlation with G3 (≈ 0.0342).

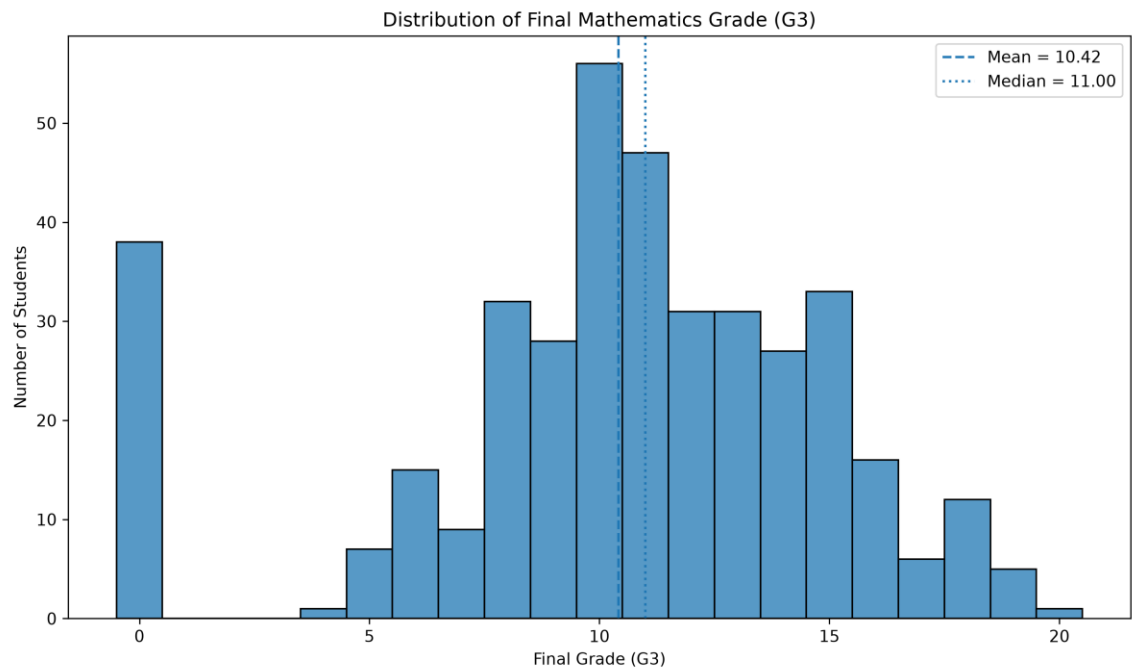


Figure 1. Distribution of final mathematics grade G3.

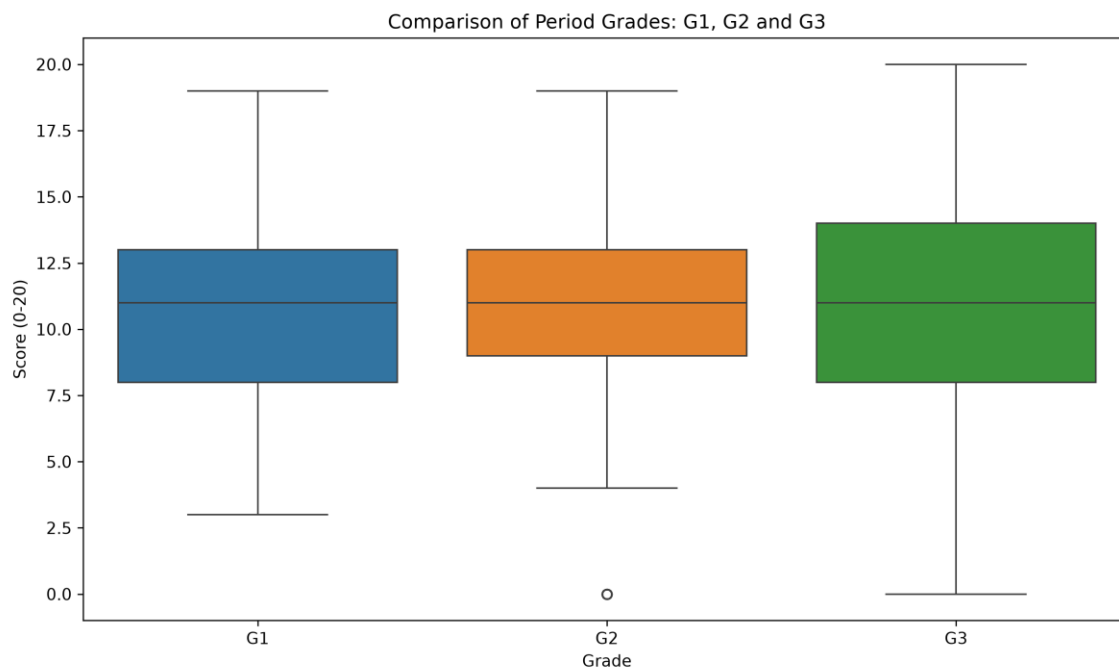


Figure 2. Comparison of G1, G2 and G3 grade distributions.

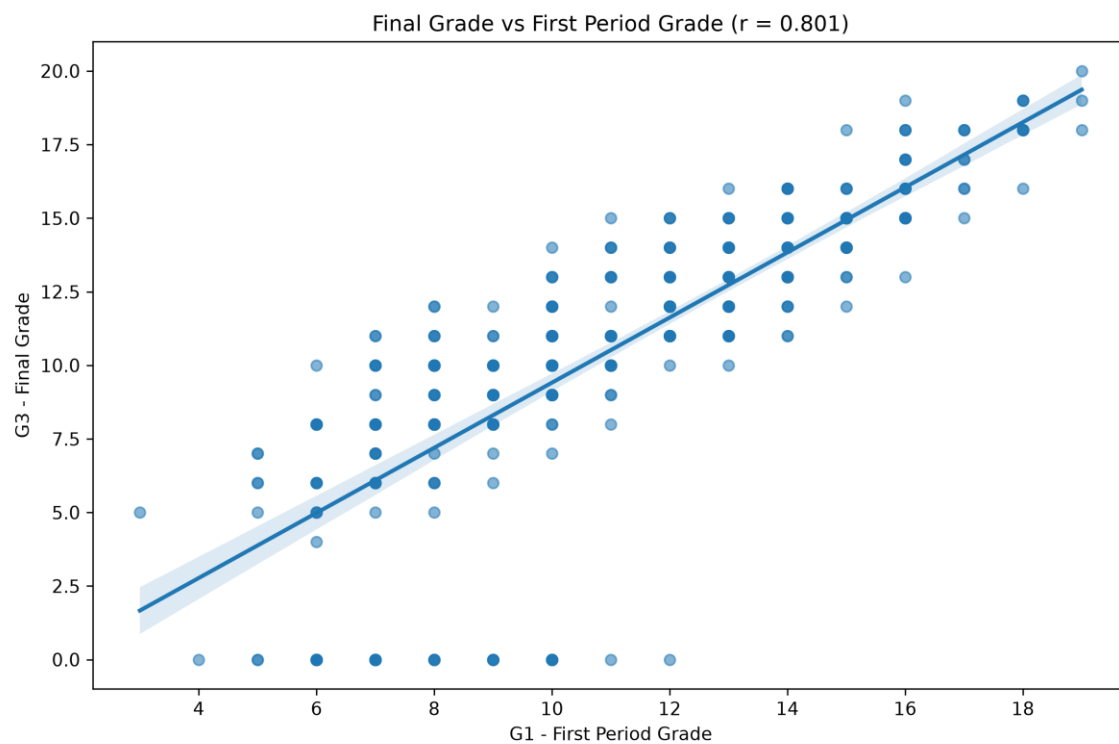


Figure 3. Relationship between first-period grade G1 and final grade G3.

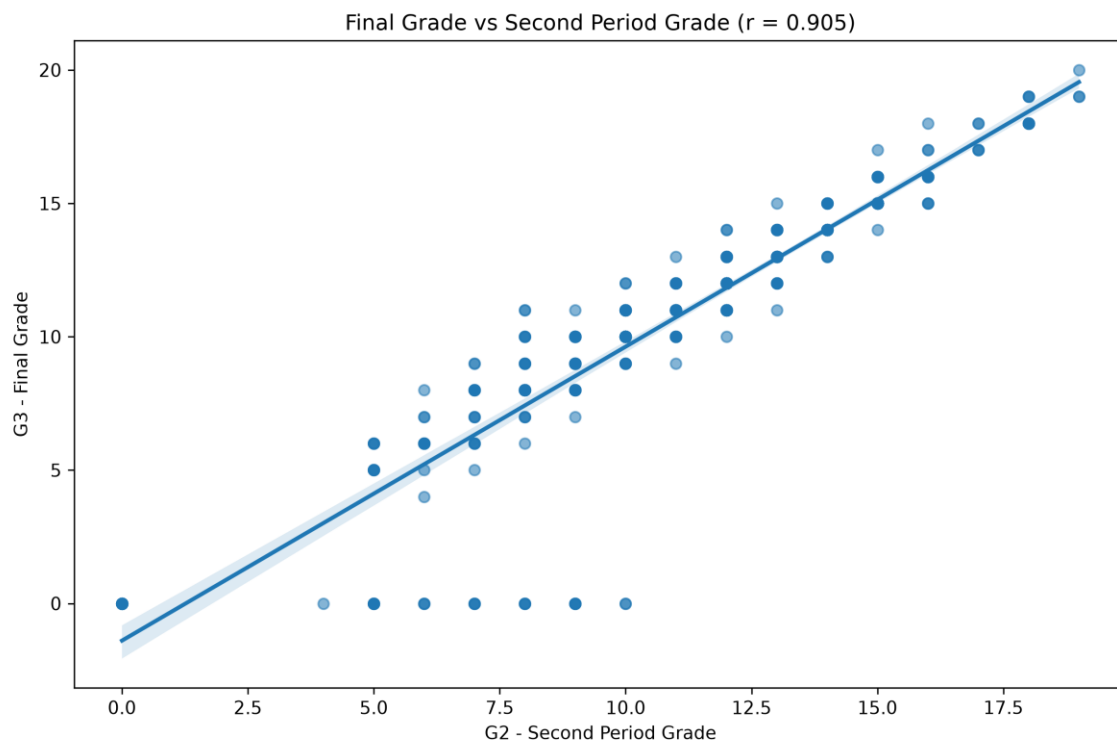


Figure 4. Relationship between second-period grade G2 and final grade G3.

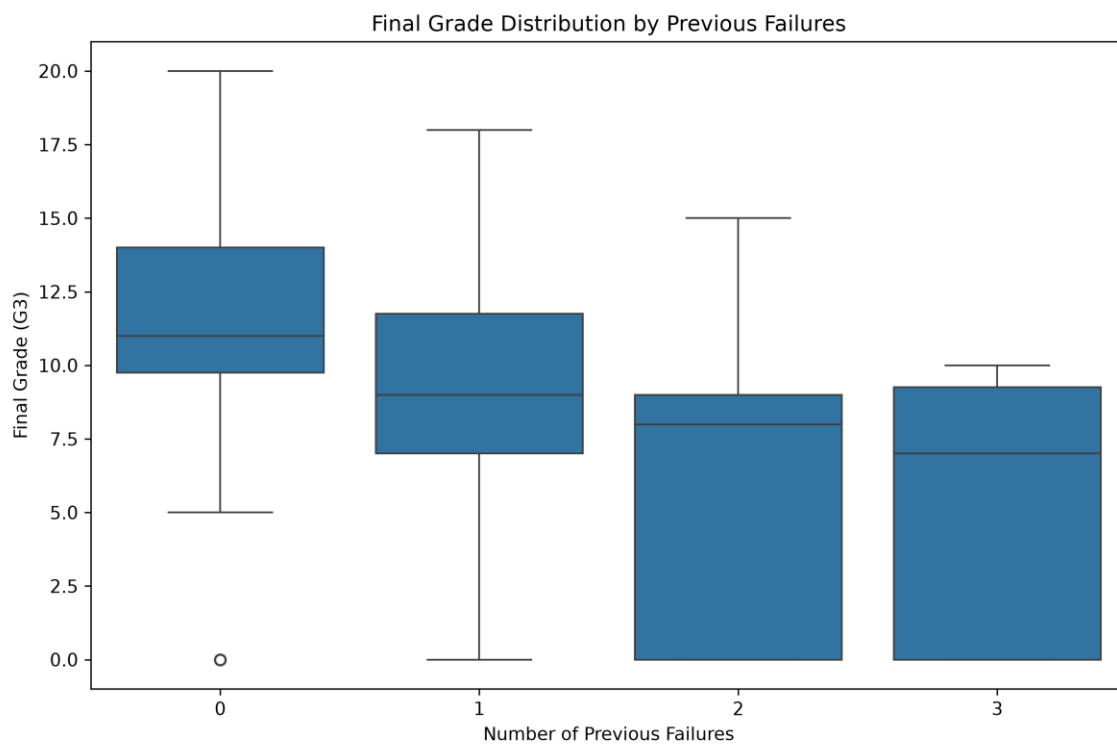


Figure 5. Final grade distribution across previous failures.

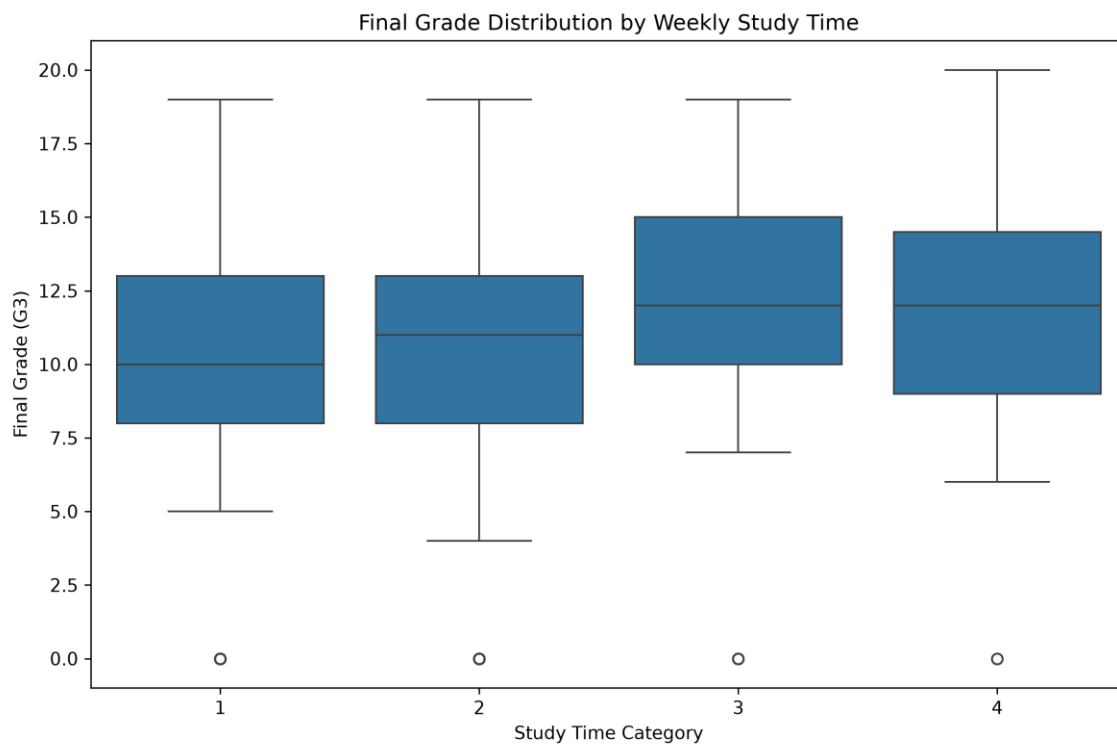


Figure 6. Final grade across study-time categories.

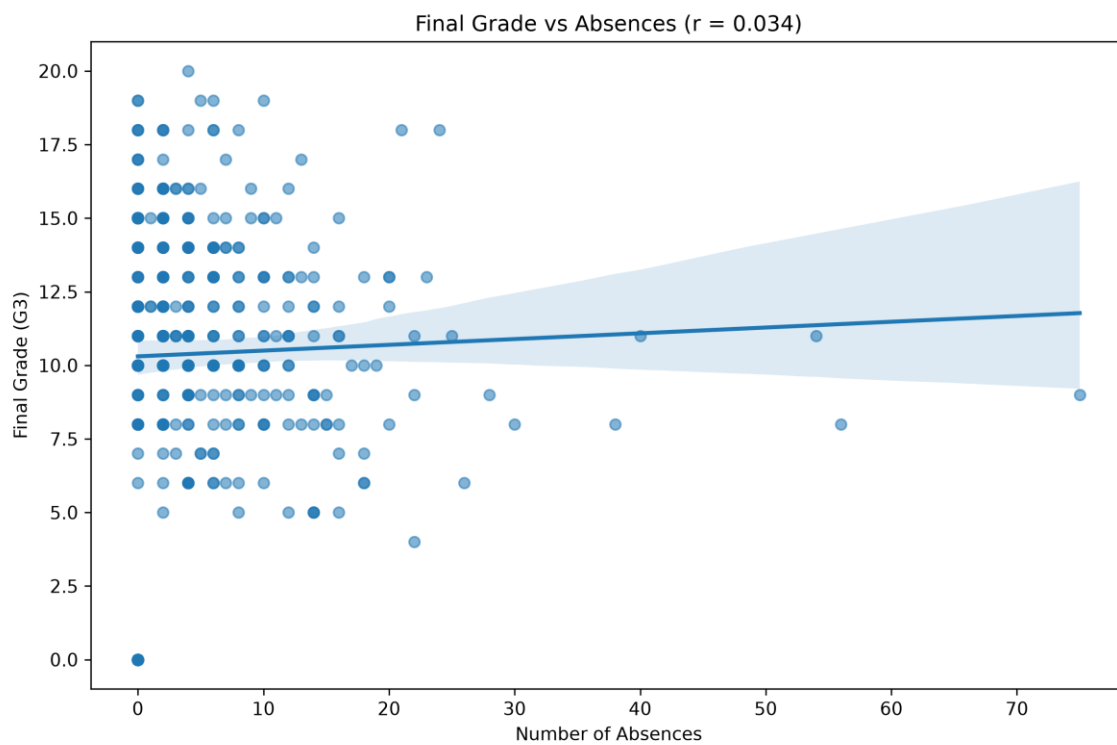


Figure 7. Relationship between absences and final grade.

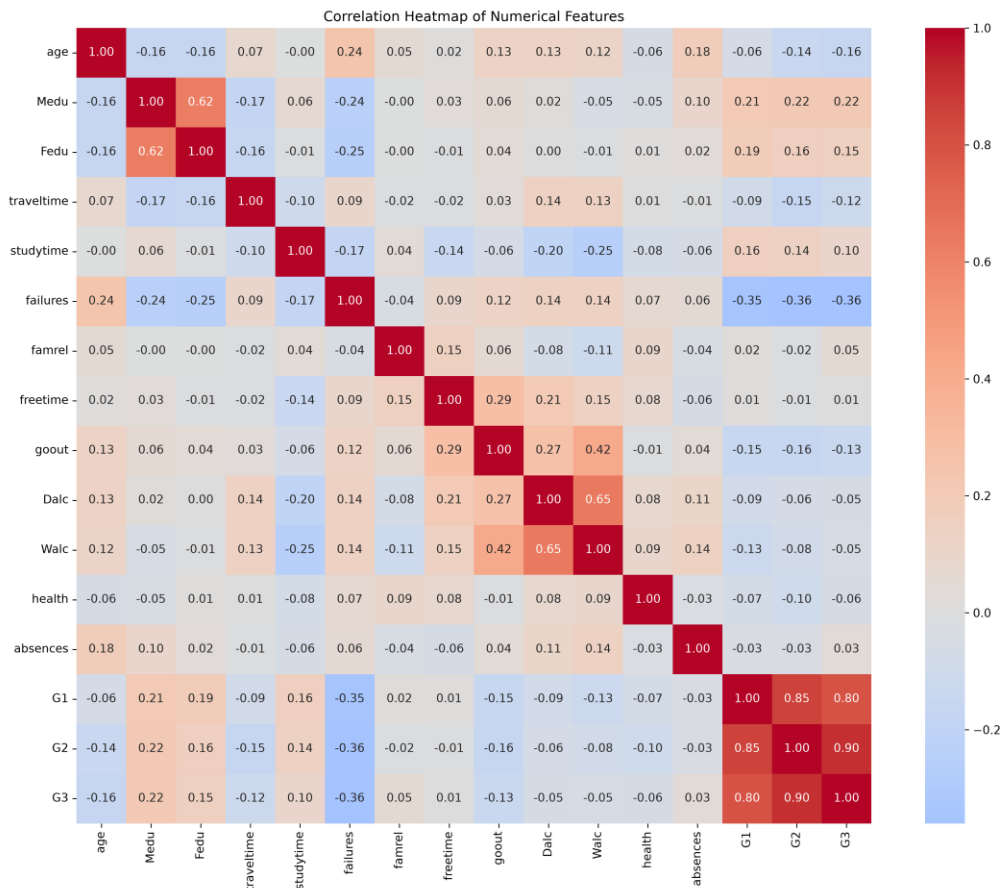


Figure 8. Correlation heatmap of numerical variables.

6. Feature Engineering and Preprocessing

Two feature scenarios were evaluated. The Full Information scenario uses all 32 non-target features, including G1 and G2. The Pre-Grade Prediction scenario excludes G1 and G2 to simulate prediction before previous-period grades are available.

Scenario	Raw Features	G1/G2 Included
Full Information	32	Yes
Pre-Grade Prediction	30	No

The data was split into training and test sets using an 80/20 split with random_state=42. Numerical features use median imputation followed by standardization. Categorical features use most-frequent imputation followed by one-hot encoding with unknown-category handling.

The preprocessing pipeline is fitted only on the training data. This prevents information from the test set from influencing preprocessing and reduces the risk of data leakage.

7. Model Development

Four supervised regression algorithms were evaluated:

- Linear Regression
- Ridge Regression

- Decision Tree Regressor
- Random Forest Regressor

The evaluation uses Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R^2 . Lower MAE, MSE and RMSE indicate smaller prediction errors, while higher R^2 indicates better explained variance.

8. Baseline Model Comparison

The baseline experiments show a clear performance difference between the Full Information and Pre-Grade Prediction scenarios. Including G1 and G2 substantially improves predictive performance.

Scenario	Model	MAE	RMSE	R^2	Rank
Full	Random Forest	1.2005	2.0150	0.8020	1
Full	Ridge	1.6391	2.3715	0.7257	2
Full	Linear Regression	1.6467	2.3784	0.7241	3
Full	Decision Tree	1.3165	2.5705	0.6778	4
Pre-Grade	Random Forest	2.9938	3.7508	0.3139	1
Pre-Grade	Ridge	3.3929	4.1934	0.1424	2
Pre-Grade	Linear Regression	3.3953	4.1957	0.1415	3
Pre-Grade	Decision Tree	3.6329	4.8379	-0.1414	4

The Full Information Random Forest achieved the best overall test performance. The Decision Tree reached a training R^2 of 1.0000, while its test performance was lower, indicating substantial overfitting.

9. Impact of Previous Grades

Removing G1 and G2 causes a substantial reduction in predictive performance. The baseline Random Forest RMSE increases from 2.0150 to 3.7508 and R^2 decreases from 0.8020 to 0.3139.

This demonstrates that previous academic performance contains strong predictive information for the final grade. It also means the Full Information model should be interpreted as a prediction system used after G1 and G2 are available, rather than a purely early-stage prediction system.

10. Hyperparameter Tuning

Random Forest hyperparameters were evaluated using GridSearchCV with 5-fold cross-validation and RMSE-based scoring.

The tuning process did not improve the held-out test performance of the baseline Random Forest. Therefore, the simpler baseline configuration was retained as the final model.

Scenario	Model	Baseline RMSE	Tuned RMSE	Outcome
Full Information	Random Forest	2.0150	2.0196	Baseline retained
Pre-Grade	Random Forest	3.7508	3.9524	Baseline retained

11. Final Model Selection

The final selected model is the Full Information Random Forest Regressor with 200 estimators, random_state=42 and n_jobs=-1. The model uses the same reproducible 80/20 train-test split used throughout the baseline evaluation.

Metric	Value
Model	Random Forest Regressor
Features	32
Training rows	316
Test rows	79
MAE	1.2005
MSE	4.0601
RMSE	2.0150
R ²	0.8020
Train RMSE	0.5687
Train R ²	0.9846

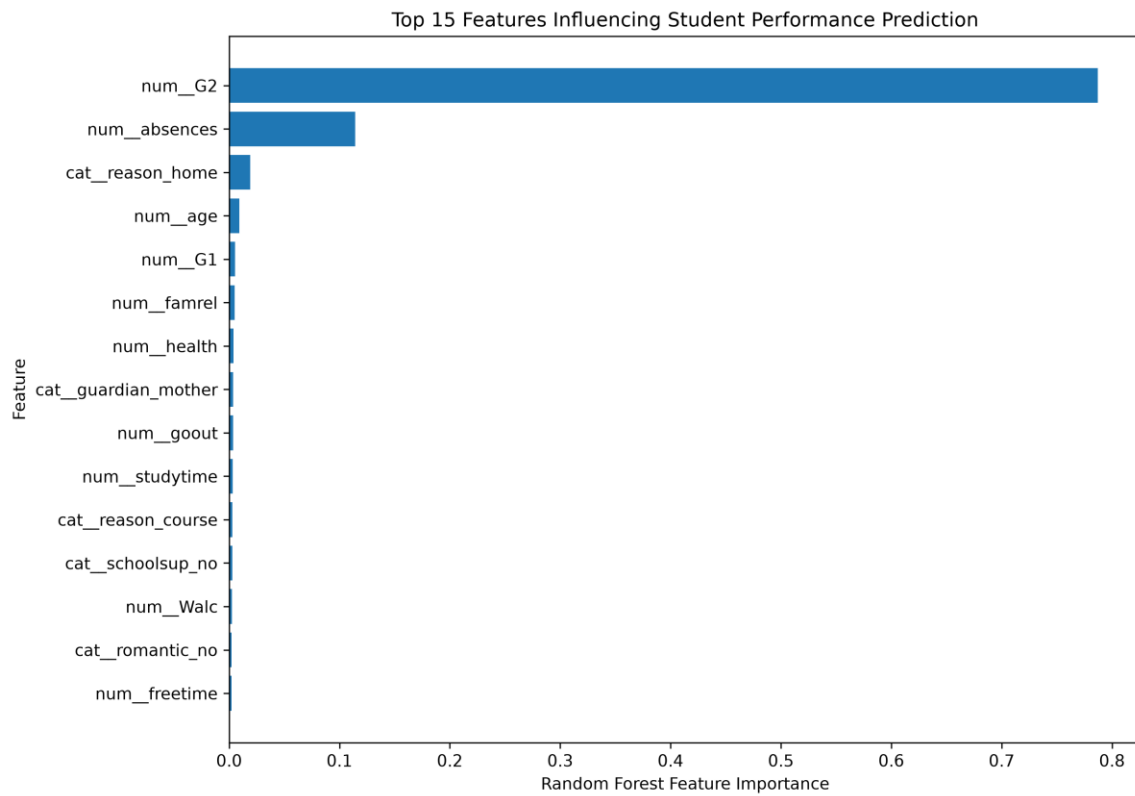


Figure 9. Final Random Forest feature importance.

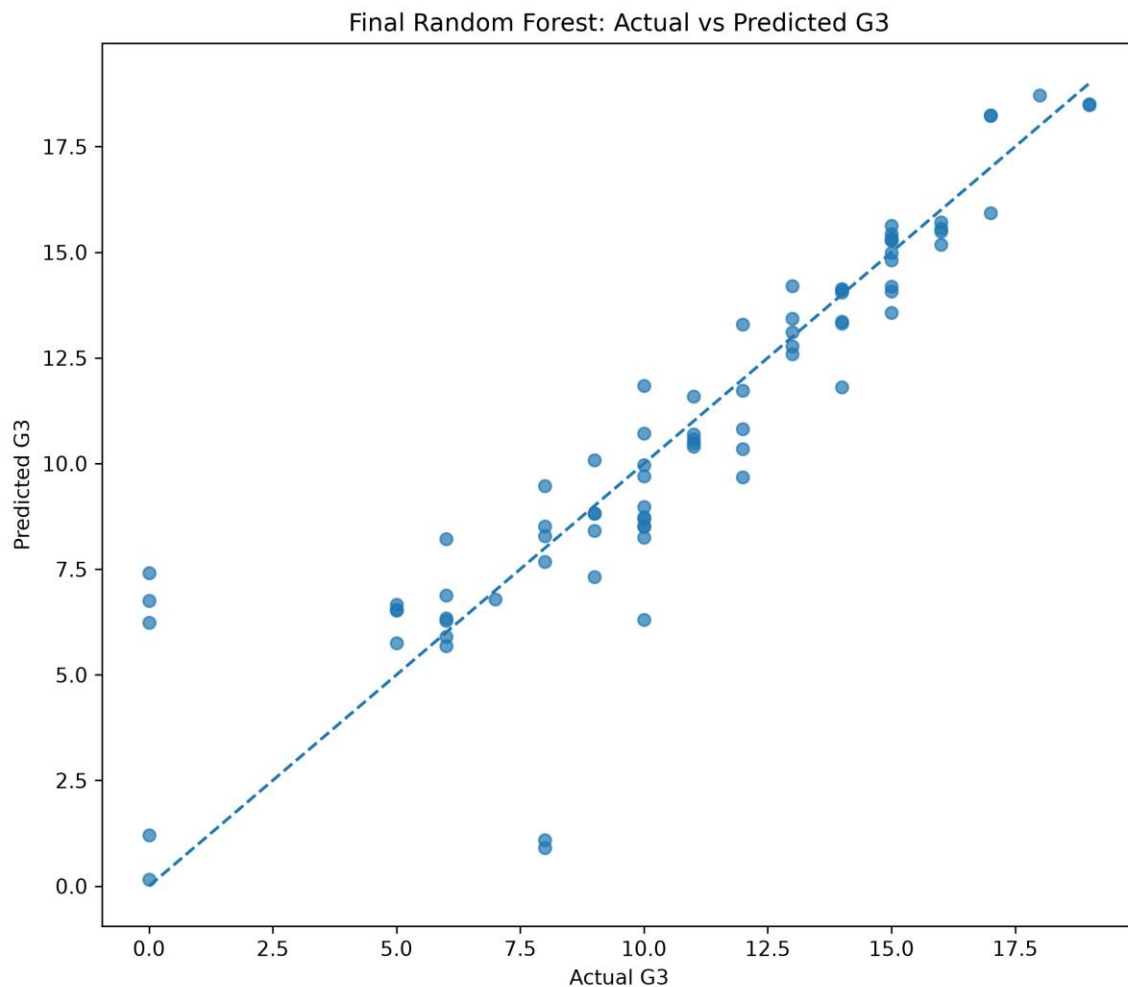


Figure 10. Actual versus predicted final grades.

12. Feature Importance Analysis

The final Random Forest assigns the largest feature importance to G2 (approximately 78.72%), followed by absences (approximately 11.43%). The remaining features contribute substantially smaller shares.

Feature	Importance
G2	78.72%
absences	11.43%
reason_home	1.90%
age	0.91%
G1	0.55%
famrel	0.50%
health	0.41%
guardian_mother	0.37%
goout	0.36%
studytime	0.32%

Feature importance describes the contribution of variables to this model's predictions; it should not be interpreted as proof that a feature causes the outcome.

13. Error Analysis

The final model has a mean prediction error of approximately 0.0373, which is close to zero overall. However, aggregate error alone does not capture the model's behavior across different grade ranges.

Actual G3 Range	Count	MAE	Mean Error	RMSE
0-5	9	3.0278	-3.0278	4.0687
6-10	29	1.3652	0.7010	2.2031
11-15	31	0.6668	0.2877	0.8901
16-20	10	0.7330	0.0950	0.8027

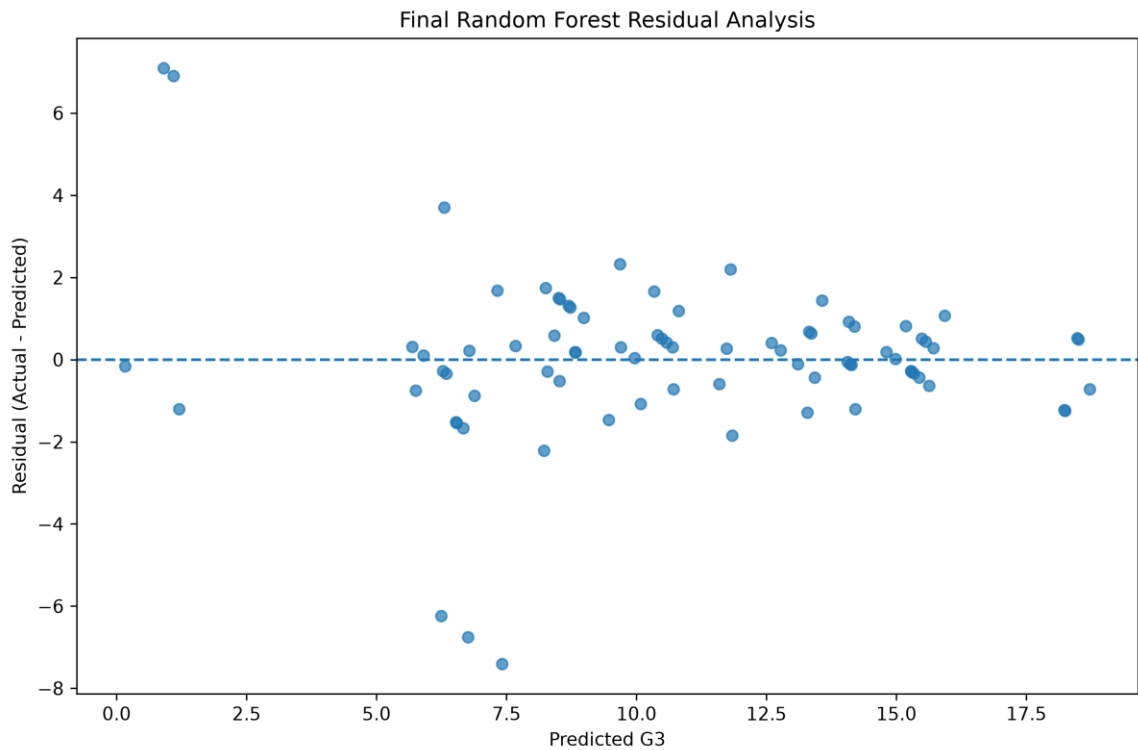


Figure 11. Residual analysis for the final model.

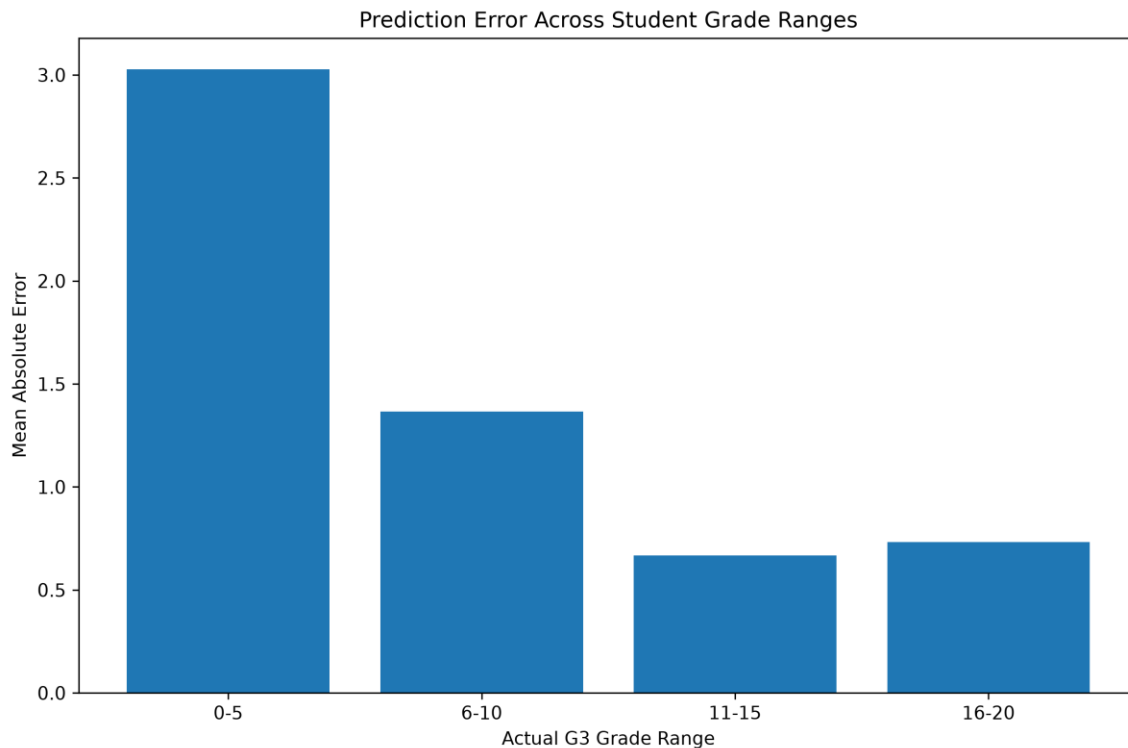


Figure 12. Prediction error across actual grade ranges.

The largest weakness is the 0–5 grade range, where the model has an MAE of 3.0278 and a negative mean error of -3.0278. This indicates that the model tends to overpredict students with very low final grades.

14. Key Findings

- The dataset contains 395 students and 33 variables with no missing values or duplicate rows.
- G1 and especially G2 have strong relationships with the final grade G3.
- Failures are negatively associated with final performance.
- Extreme absence values exist but do not provide sufficient evidence of data-entry errors and were retained.
- Random Forest outperformed Linear Regression, Ridge Regression and Decision Tree models in the Full Information scenario.
- Removing G1 and G2 substantially reduces prediction performance.
- Hyperparameter tuning did not improve the baseline Random Forest on the held-out test set.
- G2 is the dominant feature in the final Random Forest.
- The final model performs best for students with actual grades from 11 to 20.
- Very low grades are the main source of prediction error.

15. Limitations

- The mathematics dataset contains only 395 observations.
- The Full Information model relies heavily on previous grades, especially G2.
- Prediction performance decreases substantially when previous grades are unavailable.
- The model has higher error for students with very low final grades.
- Feature importance reflects model behavior and does not establish causation.

- The dataset represents students from specific Portuguese schools, so generalization to other educational populations should be treated cautiously.

16. Future Improvements

- Evaluate additional ensemble and gradient-boosting regression algorithms.
- Use repeated cross-validation to obtain more stable estimates of model performance.
- Investigate methods specifically designed to improve predictions for very low grades.
- Explore target transformation or alternative modeling strategies where justified.
- Evaluate the model on an external student dataset to assess generalization.
- Add explainability techniques such as permutation importance or SHAP for deeper interpretation.
- If early intervention is the intended use case, develop a dedicated model that excludes G1 and G2.

17. Conclusion

This project demonstrates a complete and reproducible supervised machine learning workflow for predicting final student mathematics performance. The workflow progressed from dataset verification and cleaning through exploratory analysis, preprocessing, baseline model comparison, hyperparameter tuning, final model selection and detailed error analysis.

The Full Information Random Forest was selected as the final model and achieved an RMSE of 2.0150 and R^2 of 0.8020 on the held-out test set. The results show that previous academic performance, particularly G2, provides substantial predictive information.

At the same time, the error analysis demonstrates that model performance is not uniform across students. Very low final grades remain difficult to predict accurately, providing an important direction for future model development.

18. References

- UCI Machine Learning Repository — Student Performance Dataset.
- Pedregosa et al. — Scikit-learn: Machine Learning in Python.
- McKinney — Python for Data Analysis.
- Matplotlib Documentation.
- Seaborn Documentation.
- Python Documentation.